

Data Visualization and Storytelling Report

Introduction

Data visualization is an important part of data analysis that helps in understanding patterns and trends in a dataset. In this project, different visualization techniques are used to explore the dataset and extract meaningful insights. The goal is to represent data visually and explain the findings in a simple and clear manner.

Dataset Overview

The dataset contains both numerical and categorical features. The numerical features include age, experience, and salary. The categorical features include gender, department, city, and education. This mix of data types allows different visualization techniques to be applied for better understanding.

Visualizations Used

A histogram is used to analyze the distribution of salary. It helps to understand how salaries are spread across employees and shows that most values lie in the mid-range.

A bar chart is used to compare the number of employees in different departments. It shows the distribution of employees across departments and highlights which departments have higher representation.

A scatter plot is used to analyze the relationship between experience and salary. It shows a positive relationship, meaning that salary tends to increase with experience.

A heatmap is used to study the correlation between numerical features. It helps identify relationships between variables, especially showing a strong correlation between experience and salary.

Key Insights

The analysis shows that salary increases with experience, indicating a positive relationship between the two. It is also observed that certain departments such as IT and Finance have more employees compared to others. The salary distribution shows that most employees fall into the mid-salary range, with fewer in the higher salary range. Education level also has an impact on salary variations.

Conclusion

Data visualization helps convert raw data into meaningful insights. By using different types of plots, it becomes easier to understand patterns in salary, experience, and department distribution. This improves data interpretation and supports better decision-making.